

NOVA.AI

AI PRODUCT DOCUMENTATION

AI Research & Product

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1. Product Overview

Nova Assistant is Nova.ai's flagship enterprise product — a Retrieval-Augmented Generation (RAG) based knowledge assistant that lets employees and customer-support teams query internal documents (policies, manuals, tickets, wikis) in natural language and receive grounded, citation-backed answers. It is offered as a multi-tenant SaaS product and as a single-tenant/on-premise deployment for regulated customers.

2. Core Capabilities

- **Multilingual Q&A** — supports query and response generation in English, Hindi, and 6 additional languages via automatic language detection.
- **Hybrid Retrieval** — combines BM25 keyword search with dense vector similarity, followed by cross-encoder reranking of the top-K candidates.
- **Source Citation** — every generated answer links back to the specific document and section it was grounded in, reducing hallucination risk for compliance-sensitive customers.
- **Conversation Memory** — maintains session-level context for follow-up questions without requiring the user to restate prior context.
- **Admin Analytics Dashboard** — gives customer administrators visibility into query volume, unanswered queries, and document coverage gaps.

3. High-Level Architecture

User Query → API Gateway → NLP Preprocessing (language detection, query rewriting) → Hybrid Retriever (BM25 + Vector DB) → Reranker (Cross-Encoder) → Prompt Orchestration (Jinja templates) → LLM Inference (via OpenRouter multi-model routing) → Post-processing & Citation Mapping → Response.

4. Supported Data Sources

Source Type	Supported Formats	Update Frequency
Document repositories	PDF, DOCX, TXT, Markdown	Real-time / on upload
Spreadsheets	XLSX, CSV	Real-time / on upload
Ticketing systems	Zendesk, Freshdesk (via connector)	Every 15 minutes
Internal wikis	Confluence, Notion (via connector)	Every 30 minutes
Chat transcripts	Slack, Teams (via connector, opt-in)	Every 15 minutes

5. Model & Provider Strategy

Nova Assistant routes inference requests through OpenRouter, allowing customers to select from multiple underlying LLM providers based on cost, latency, and data-residency requirements. Embeddings default to a multilingual BGE-family model, benchmarked against E5 variants during onboarding to select the best fit for the customer's document corpus.

Data Handling Commitment

Customer documents are never used to train underlying foundation models. All customer data is encrypted at rest and in transit, logically isolated per tenant in the multi-tenant deployment, and fully deletable within 30 days of contract termination, consistent with the Data Classification Policy.

6. Evaluation & Quality Metrics

Each customer deployment is benchmarked before go-live and monitored continuously against Recall@K, Mean Reciprocal Rank (MRR), and a Faithfulness score (measuring whether the generated answer is fully supported by retrieved context). Deployments falling below the agreed Faithfulness threshold trigger an automatic retrieval-pipeline review by the AI Research team.

7. Latency & Availability SLAs

Metric	Target	Measured As
p95 response latency	< 3.5 seconds	End-to-end, query to answer
Platform availability	99.9% monthly uptime	Excluding planned maintenance
Retrieval index freshness	< 15 minutes	For connector-based sources

8. Customization & Configuration

Customers can configure chunking strategy (fixed, recursive, semantic, or document-aware), response tone and length via prompt templates, and role-based answer restrictions so that certain document sets are only queryable by authorized user groups, aligned to the Data Classification Policy's access-control principles.

9. Deployment Options

Nova Assistant is available as a multi-tenant SaaS offering (fastest time-to-value), a single-tenant cloud deployment (dedicated infrastructure, same cloud region as the customer), or an on-premise/VPC deployment for customers with strict data-residency or air-gapped requirements, most commonly selected by BFSI and government customers.

10. Related Documents

[System Architecture Guide](#) · [API Documentation](#) · [Information Security Guidelines](#) · [Data Classification Policy](#) · [Customer Support Manual](#).

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